SHIV NADAR
UNIVERSITY
CHENNAI

Analysis of Whether Credit Card Usage Has Led to Debt Accumulation Among Different Age Groups

DISSERTATION

Under the Guidance of

Dr. Rohit M– Associate Professor

C ANIRUDH

(22031101033)

BSc. Economics (Data Science)

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Declaration

I hereby declare that the **Dissertation** entitled **"Analysis of Whether Credit Card Usage Has Led to Debt Accumulation Among Different Age Groups"** is almost entirely my own research work. Any text, data, or images resulting from the work of others (whether published or unpublished) are fully acknowledged in the text, bibliography or in footnotes.

(C Anirudh)

Forwarded by

(Dr. Rohit M)

Evaluated by

(Supervisor) (Examiner 1) (Examiner 2) (External Examiner)

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Introduction

The introduction of credit cards has revolutionized personal finance, fundamentally changing how individuals manage and allocate their resources. Credit cards provide instant purchasing power without an immediate cash outflow, offering convenience and flexibility in transactions. They also come with benefits such as rewards, cashback incentives, travel perks, and the ability to manage large expenses through installment plans. These features have made credit cards indispensable financial tools for many consumers. However, despite their advantages, credit cards are not without challenges. Easy access to credit can lead to pitfalls like overspending, accumulating high-interest debt, and fostering financial mismanagement. Behavioural economics provides a useful lens for understanding these patterns, as cognitive biases and preferences (e.g. present bias, reward-seeking) can influence how people use credit and handle debt.

Notably, credit card usage and its consequences vary significantly across different age groups. Key demographic factors such as age, income level, lifestyle, and financial experience influence patterns of credit card use. Younger adults (for example, those in their 20s) often view credit cards as tools for convenience and are drawn to reward programs offering cashback or discounts. In contrast, individuals in their 30s may prioritize managing household expenses and leverage credit cards for larger financial commitments, while those in their 40s tend to focus on financial stability and credit-building, often demonstrating more cautious and deliberate. Banks and financial institutions play a critical role in shaping these behaviours through marketing tactics that encourage greater adoption and usage. Enticing reward schemes, promotional discounts, and flexible repayment options (like Equated Monthly Installments, EMIs) provide tangible benefits but may also promote overspending and long-term dependency on credit, exploiting cognitive biases such as underestimating future expenses or favoring immediate gratification. This intersection of institutional strategies and individual behaviour is central to understanding credit card usage outcomes.

The purpose of this research is to provide a comprehensive analysis of how credit card usage may lead to debt accumulation across different age groups. We examine whether usage behaviour differs by age and how these differences contribute to the likelihood of credit card debt. We investigate the relationship between age bracket (20–30, 30–40, 40–50 years), credit card usage patterns, and the incidence of debt accumulation. By combining quantitative analysis of survey data with qualitative insights from open-ended responses, the study aims to shed light on age-specific credit card behaviours and their financial consequences. The overarching research question is: **Does credit card usage influence spending behaviour and debt accumulation differently across age groups, and if so, how?** Answering this question can inform targeted financial education and credit management strategies for different demographic segments.

Review of Literature

Credit cards offer unparalleled convenience in payments, but their proliferation has introduced concerns regarding overspending and debt accumulation. Prior studies have explored these dynamics across various populations, providing context for the age-wise patterns investigated in this paper. In the Indian context, research highlights that credit card usage patterns differ markedly by age group. **Khare (2015)** observed that younger consumers in India are more likely to own and use credit cards compared to older consumers, a trend attributed to materialistic values and the social status associated with credit card ownership among youth. Similarly, a survey by **(Allirajan, 2017)** reported that 24% of young Indian adults aged 18–24 prefer credit cards as their primary payment method to avoid carrying cash, whereas 33% of adults aged 45 and above use credit cards mainly for the convenience of "buy now, pay later". These findings suggest that younger users may embrace credit for its convenience and perks, while older users employ it as a financial management tool, potentially influencing how debt is accumulated.

Banks and financial institutions have been found to play a pivotal role in shaping credit card usage through targeted marketing strategies. **Sharma et al. (2023)** analyzed the effectiveness of various marketing approaches (promotional offers, rewards programs, customer service enhancements) on credit card adoption and usage. Their findings indicate that rewards programs and promotions are particularly successful in attracting and retaining customers across demographics, though factors such as age, income, and education level significantly influence the effectiveness of these strategies. For instance, younger cardholders may be more enticed by cashback and discounts on trendy purchases, while older customers might respond to promotions on insurance or medical benefits. These institutional strategies can inadvertently encourage higher spending; regulators have taken note of such trends. In October 2024, Reserve Bank of India Governor Shaktikanta Das cautioned non-banking financial companies (NBFCs) against adopting a "growth-at-any-cost" strategy for credit issuance. He warned that chasing aggressive growth through high interest rates and fees can lead to imprudent lending and rising consumer indebtedness, posing risks to financial stability. This regulatory perspective underscores the broader economic concern that unchecked credit promotion might contribute to debt problems among consumers.

The relationship between credit card usage and consumer spending behaviour is complex and has been examined in behavioural studies. **Khare (2016)** explored how money attitudes, credit card usage, age, and gender affect compulsive buying behaviour among Indian consumers. The study found that compulsive buying is driven largely by attitudes toward money—especially feelings of power and price sensitivity—with age acting as a moderating factor. Younger consumers, due to perceived empowerment from credit cards (the ability to purchase beyond current cash means), may be more vulnerable to overspending and subsequent debt, whereas older consumers might exercise more restraint. This is consistent with findings from other contexts that suggest younger adults often accumulate higher credit card debt and experience more stress related to that debt, whereas middle-aged and older adults might manage credit more prudently. Additionally, rewards programs have a notable influence on spending: a recent **(Chelani, 2024)** survey reported that nearly 60% of respondents prioritize reward benefits

when selecting a new credit card. Such incentives can encourage increased usage — for example, spending more to earn cashback or airline miles — which might inadvertently lead to higher balances if not paid off promptly.

In summary, existing literature suggests that younger consumers are often more enthusiastic users of credit cards but are at risk of poorer financial outcomes like debt accumulation, while older consumers typically use credit cards more conservatively. Institutional incentives (rewards, promotions) and personal attitudes (materialism, financial literacy) both influence these behaviours. These insights from prior studies set the stage for our analysis, which empirically examines how different age groups in our survey sample use credit cards and how those behaviours correlate with debt incidence. Our study extends this body of literature by combining behavioural data with sentiment analysis and predictive modelling to understand not just *if* age groups differ in credit outcomes, but *why*—considering psychological factors and usage patterns behind the numbers.

Methodology

This study adopts a mixed-methods approach, integrating quantitative analysis with textual data mining, to explore the relationship between age group, credit card usage patterns, and debt accumulation. The research process involved several stages: data collection via survey, data preparation, exploratory analysis, statistical testing, sentiment analysis, clustering, and predictive modelling. Each stage is described below.

Data Collection: This study was done using a structured survey to gather primary data on credit card usage and its perceived influence on spending behaviour across three age brackets: 20–30, 30–40, and 40–50 years. A total of 136 respondents (recruited online via convenience sampling) participated, ensuring a substantial representation of each age group. The survey questionnaire included both close-ended and open-ended questions. Close-ended items covered demographics (age group), credit card ownership status, number of cards owned, usage frequency (daily/weekly/monthly/rarely), typical spending patterns, repayment behaviour (e.g., whether the user usually pays the balance in full each month), experience with rewards programs and EMIs (Equated Monthly Installments), and whether the individual has ever faced credit card debt issues. An example close-ended question is *"Do you usually pay your credit card balance in full? (Yes – Always / Sometimes / Rarely / Never)"*. Open-ended questions were included to capture qualitative insights, for instance: *"Do you feel that using a credit card influences your spending behaviour? If so, how?"*. This open-ended prompt allowed respondents to elaborate on the psychological or behavioural impact of credit card usage in their own words. The mix of question types was intended to provide both quantifiable metrics and richer context regarding credit habits.

Data Preparation and Cleaning: The survey responses were exported to a spreadsheet and then analyzed using Python (Pandas library). Several preprocessing steps were used to prepare the data for analysis. First, column names were standardized for consistency and ease of reference. Missing or invalid entries were handled by removing or correcting responses that were incomplete or not relevant (for example, a few respondents entered "NA" or left text responses blank; these were treated as missing). Categorical variables such as Age Group,

Usage Frequency, Full Payment Behaviour, and Debt Status (whether the person had faced credit card debt issues) were label-encoded for analysis. For instance, Age Group was encoded as ordinal categories (1, 2, 3 corresponding to 20–30, 30–40, 40–50 respectively), and the binary outcome "Faced Credit Card Debt (Yes/No)" was encoded as 1/0. Open-ended text responses were cleaned by removing irrelevant or non-informative answers (some respondents simply answered "No" or "Not really" to the open-ended question; such responses were filtered out to focus on those who provided substantive feedback). This cleaning ensured that the qualitative data used in sentiment and clustering analysis was meaningful.

Exploratory Data Analysis (EDA): Computed descriptive statistics and generated visualizations to understand the distribution of responses and key trends in the data. This included frequency counts and percentages for major categorical variables (ownership, number of cards, usage frequency, payment behaviour, etc.) overall and by age group. We examined the distribution of credit card challenges and preferences (such as preferred reward types) across ages to see if certain age groups exhibit distinct preferences (e.g., younger users favouring cashback vs. older users favouring travel rewards). Cross-tabulations between age group and other variables were created to spot potential associations (for instance, age group vs. "pays balance in full" or age group vs. "faced debt issues"). We also constructed visualization aids like bar charts and histograms. A correlation matrix was computed (after encoding categorical responses numerically) to assess relationships among variables like age, usage frequency, repayment behaviour, and debt outcome. This was visualized as a heatmap for clarity. Pair plots were also generated to visualize pairwise relationships between key variables (Age, Usage Frequency, Payment behaviour, and Debt) to identify any noticeable patterns or clusters in the data. The EDA was crucial for formulating initial hypotheses (e.g., seeing that one age group might have disproportionately high debt incidence) and informed the subsequent statistical tests and modelling.

Statistical Test – Chi-Square Analysis: Based on EDA observations, we conducted a Chi-square test of independence to formally assess whether age group and credit card debt incidence are statistically independent or associated. A contingency table was created of Age Group (3 categories) vs. Debt Outcome (Yes/No) and computed the Chi-square statistic, corresponding p-value, and expected frequencies under the null hypothesis of no association. This test helps determine if any age group is significantly more likely to have experienced debt issues than would be expected by chance. A significance level of 0.05 was used for hypothesis testing. The Chi-square result provides a quantitative underpinning to the observed differences in debt occurrence among age groups.

Sentiment Analysis: To quantify the qualitative feedback from the open-ended question about spending behaviour influence, sentiment analysis was applied. Using the TextBlob library (a Python NLP toolkit), each open-ended response was evaluated for sentiment polarity, which is a score ranging from –1 (extremely negative sentiment) to +1 (extremely positive sentiment). For example, a response like *"Yes, I often overspend with credit cards and regret it later"* would yield a negative polarity, whereas *"Credit cards are convenient and haven't caused me any issues"* might yield a neutral to slightly positive polarity. The sentiment score was calculated

for each respondent's comment and then examined the distribution of these scores. The average sentiment score was computed for each age group to see if younger or older respondents differed in how positively or negatively, they viewed the influence of credit cards on their spending. Sentiment results were visualised using histograms and density plots to see the overall skew (e.g., whether many comments were negative) and bar charts for average sentiment by age group (to compare group tendencies). This text analysis adds a psychological dimension to the study, indicating whether those who have experienced or fear debt express more negative sentiment, and if such expressions align with age-related trends.

Clustering of Open-Ended Responses: Beyond sentiment polarity, an unsupervised clustering was performed on the textual responses to identify common themes or behaviour patterns described by the respondents. Then,Converted the cleaned open-ended responses into a numerical form using TF-IDF vectorization (Term Frequency–Inverse Document Frequency), which represents each response in a high-dimensional space based on the importance of words used (down weighting extremely common words and emphasizing more unique terms). We then applied K-Means clustering to group the responses into a few clusters based on similarity. After experimenting with different cluster counts, we settled on four clusters that provided distinct thematic groupings of responses. To interpret these clusters, we examined the top contributing words/phrases in each cluster and read a few representative responses from each cluster. Principal Component Analysis (PCA) was used for dimensionality reduction to visualize the clusters in two dimensions for understanding (though the visualization is primarily for exploratory insight rather than included as a result figure). This clustering aimed to uncover latent groupings such as: respondents who feel in control of their spending, those who admit to overspending, those who use credit strategically for rewards, etc. The clusters derived were then qualitatively labelled according to their central theme.

Predictive Modeling: Finally, we approached the question of debt accumulation from a predictive analytics standpoint. We framed "facing credit card debt issues" as a binary outcome (Yes = 1, No = 0) to be predicted from a set of predictor variables. The predictors considered were the key behavioural factors: age group, usage frequency, and full payment behaviour. These were chosen based on both intuition and earlier analysis suggesting they are related to debt incidence. We split the data into training and testing sets and trained three classification models using scikit-learn: (1) **Logistic Regression**, as a baseline linear model that provides interpretable coefficients; (2) **Support Vector Machine (SVM)**, with both linear and non-linear (RBF) kernels to capture more complex decision boundaries; (3) **Random Forest Classifier**, an ensemble tree-based model that can capture non-linear relationships and highlight feature importance. Model performance was evaluated primarily by accuracy (overall percentage of correct predictions) and precision/recall for the positive class (debt = Yes), since the classes were imbalanced (only 21% of respondents had faced debt issues). We especially focused on the recall for the debt class, as a key interest was whether the model could correctly identify individuals at risk of debt. We computed the confusion matrix for each model and derived metrics including the F1-score. In addition, for the logistic model, we examined the coefficients to understand the direction and strength of influence of each predictor on debt risk (for instance, whether being older or always paying in full reduces the probability of debt, as hypothesized). We also explored techniques to address class imbalance, such as resampling, but

given the relatively small dataset (and issues with applying certain oversampling methods like SMOTE on this data), we present the unadjusted results with careful interpretation of the limitations. The combination of these methods provides a robust analysis: the statistical test establishes if age and debt are related, sentiment and clustering provide behavioural context, and the predictive modeling evaluates how well we can anticipate debt occurrences from observable traits.

By employing this multi-faceted methodology, we ensure that the analysis not only tests the core hypothesis (differences in debt outcomes by age group) but also digs deeper into underlying behaviours and perceptions that might explain those differences. In the next sections, we describe the data and discuss the results of these analytical steps.

Data Description

The dataset consists of **136 survey responses** from individuals in the specified age groups of 20–30, 30–40, and 40–50. The sample is relatively balanced among these categories, with 20–30 and 30–40 each contributing 41 respondents, and the 40–50 group slightly higher at 54 respondents. All participants were asked a uniform set of questions about their credit card usage and experiences. Key variables in the dataset include:

- **Age Group:** Categorical (20–30, 30–40, 40–50). Each respondent self-identified into one of these brackets. As noted, the distribution is roughly even, which aids comparative analysis without heavy weighting for any single group.
- **Credit Card Ownership:** Binary (Yes/No). Nearly all respondents (129 out of 136) reported currently owning at least one credit card, which indicates that the survey effectively targeted credit card users. This high ownership rate means subsequent questions about card usage are applicable to the vast majority of the sample.
- **Number of Credit Cards:** Categorical (e.g., “1–2”, “3–4”, “5+”). This gauges how many cards each person holds. A preliminary tally shows that most respondents have *1–2 cards* (the most common category), suggesting a preference for either keeping the number of cards low or being early in one’s credit journey. Fewer respondents have many cards (very few reported holding 5 or more cards).
- **Usage Frequency:** Categorical with ordered categories (Daily, Weekly, Monthly, Rarely). This measures how often individuals use *any* of their credit cards. It serves as a proxy for dependence on credit for routine transactions. For example, “Daily” users likely rely on credit cards for everyday purchases, whereas “Rarely” might indicate use only for emergencies or specific needs.
- **Primary Purpose of Credit Card Use:** Categorical (multiple choice). Respondents indicated their main reasons for using credit cards (such as daily expenses, large purchases, travel, emergencies, etc.). This variable provides context on whether cards are used as a convenience tool, for leveraging credit on big expenses, or for perks like travel.

- **Payment Behaviour – Full Balance Payment:** Categorical (Yes – Always, Sometimes, Rarely, Never). This crucial variable captures the individual's repayment habit – whether they typically pay off their credit card bill in full each cycle or carry a balance. This is directly tied to interest accumulation and debt: those who "always" pay in full avoid interest and typically debt, whereas those who "rarely" or "never" pay in full are likely accruing debt.
- **Credit Limit Management:** We asked if respondents had ever requested a credit limit increase (Yes/No) and typical credit utilization range (<30%, 30–70%, >70%, or even exceeding limit) to understand credit management. These factors can signal risk appetite or financial discipline.
- **Experience of Penalties/Fees:** Binary question on whether they had ever incurred late payment fees or other penalties on their credit card. This can indirectly indicate instances of financial stress or oversight.
- **Faced Credit Card Debt Issues:** Binary (Yes/No). This is the target variable for our analysis – whether the respondent has ever encountered issues related to credit card debt (e.g., difficulty in payments, needing debt restructuring, etc.). In our sample, 29 out of 136 respondents (approximately 21%) answered "Yes," indicating they have faced such issues.
- **Perception Variables:** We included questions like "Do you find credit card rewards useful?" (Yes/No/Neutral) and "Would you recommend getting a credit card to others?" (Likert scale) to gauge satisfaction and perceived benefit. These can vary with experience; for instance, someone who fell into debt might respond differently about recommending credit cards than someone who has had a smooth experience.
- **Open-Ended Response on Spending Influence:** Text. As discussed, participants provided their personal view on how using a credit card influences their spending behaviour. This qualitative variable enriches our understanding of the psychological impact and is later quantified via sentiment analysis.
- **Derived Sentiment Score:** Numerical (–1 to +1). For analysis purposes, we created this variable by applying sentiment analysis to the open-ended responses. A positive score indicates a generally positive or neutral stance (e.g., feeling in control or benefiting from credit card use), whereas a negative score indicates the respondent expresses concern or negative experiences (e.g., overspending, regret).

In terms of **demographics and distribution**, the sample's age distribution has a slight skew toward the oldest bracket (40–50). Each age group shows a high prevalence of credit card ownership (virtually all participants in each group have cards, given the overall 95% ownership rate). The balance in age categories is intentional to facilitate comparisons, and the overall high incidence of card ownership and usage frequency underscores that this is a population actively engaged with credit card products.

Crucially, about **21% of respondents reported having encountered debt issues related to credit cards**. While this is a minority, it is a significant subset that allows us to compare

characteristics between those who have faced difficulties and those who have not. This dataset, although based on self-reported survey data (which may include biases such as under-reporting of negative behaviours), is rich enough to allow analysis of behavioural patterns (like spending and repayment habits) and their relationship with age and debt outcomes.

Overall, the data provides a comprehensive snapshot of credit card usage behaviour — from how often people swipe their cards to how they pay the bills, and how they emotionally perceive the impact of credit cards on their spending. In the next section, we present the analysis and results, focusing on patterns that emerge across different age groups and how these patterns relate to debt accumulation.

Analysis and Results

Exploratory Data Analysis

We begin by examining general usage patterns and behaviours in the sample, before honing in on the age-wise differences and debt outcomes. Most respondents exhibit what could be considered *responsible* credit habits on the surface: for instance, **75% of respondents indicated that they "always" pay their credit card balance in full each month**, suggesting a high level of financial discipline in avoiding interest charges. Only a small fraction reported "Rarely" or "Never" paying in full. Additionally, nearly **85 respondents (out of 136) own 1–2 credit cards**, making this the most common category for number of cards held. Fewer people have three or more cards, which could reflect either personal choice to limit credit exposure or earlier stage of credit usage (younger users might not have accumulated multiple cards yet). In terms of usage frequency, the sample skews toward regular use: the most frequent response was *Daily usage* (45 individuals), followed by *Weekly* (37). Together, about 60% of respondents use their cards at least weekly, implying that credit cards are well integrated into routine spending for most. Around 24% use them *Rarely* (33 people) or *Monthly* (21 people), indicating a minority use credit cards sparingly or only for specific kinds of purchases.

Looking at the *primary reasons for using credit cards*, two motivations stood out across the sample: everyday **expenses** and **large purchases**. Many respondents (especially in younger and middle age groups) reported using credit cards for day-to-day expenses like groceries and bills, citing convenience and rewards. A comparable number also cited using cards for large, planned expenditures (such as electronics or appliances), often to avail no-cost EMI options or to avoid dipping into savings. Some niche reasons like "travel" or "online shopping offers" were mentioned, but less frequently. It's worth noting that a handful of respondents wrote in combined reasons (e.g., "All of the above" or multiple selections), which suggests overlapping use-cases – those individuals use cards extensively for varied purposes.

Despite widespread usage, certain prudent behaviours were common. For example, **over 70% of participants indicated they have *never* requested a credit limit increase** from their bank. This could suggest that most are comfortable with their given credit limits or are cautious about not over-extending credit. In terms of credit utilization, when asked approximately what percentage of their credit limit they typically use, the majority (66 respondents) reported staying below 30% utilization – a habit often recommended to maintain a good credit score.

Only a very small number (3 respondents) admitted to regularly exceeding 100% of their limit (which likely means incurring over-limit fees), highlighting that such risky behaviour is quite rare in the group.

An interesting insight emerges from the **rewards and benefits** perspective: A significant majority (around 71 respondents) believe that credit card rewards (points, cashback, etc.) are useful and add value to their spending, whereas about 47 were neutral on this, and only a few felt rewards were not useful. This generally positive view on rewards is consistent with the high usage frequency – people who use cards often tend to value the benefits. The *most preferred reward type* among our respondents was **Cashback**, chosen by 59 people, which outpaced travel rewards (39 people) and shopping vouchers (29 people). Cashback's popularity might be due to its versatility and immediate tangible benefit, and it was especially favored by younger users looking for instant savings on purchases.

To delve deeper, we explored how some of these behaviours correlate with each other. We encoded key behavioural variables (Age as numeric 1–3, Usage frequency 0–3 from Rarely to Daily, Full payment behaviour 0–3 from Never to Always, and Debt outcome 0/1) and computed a correlation matrix. **Figure 2** below shows a heatmap of Pearson correlation coefficients between selected variables:

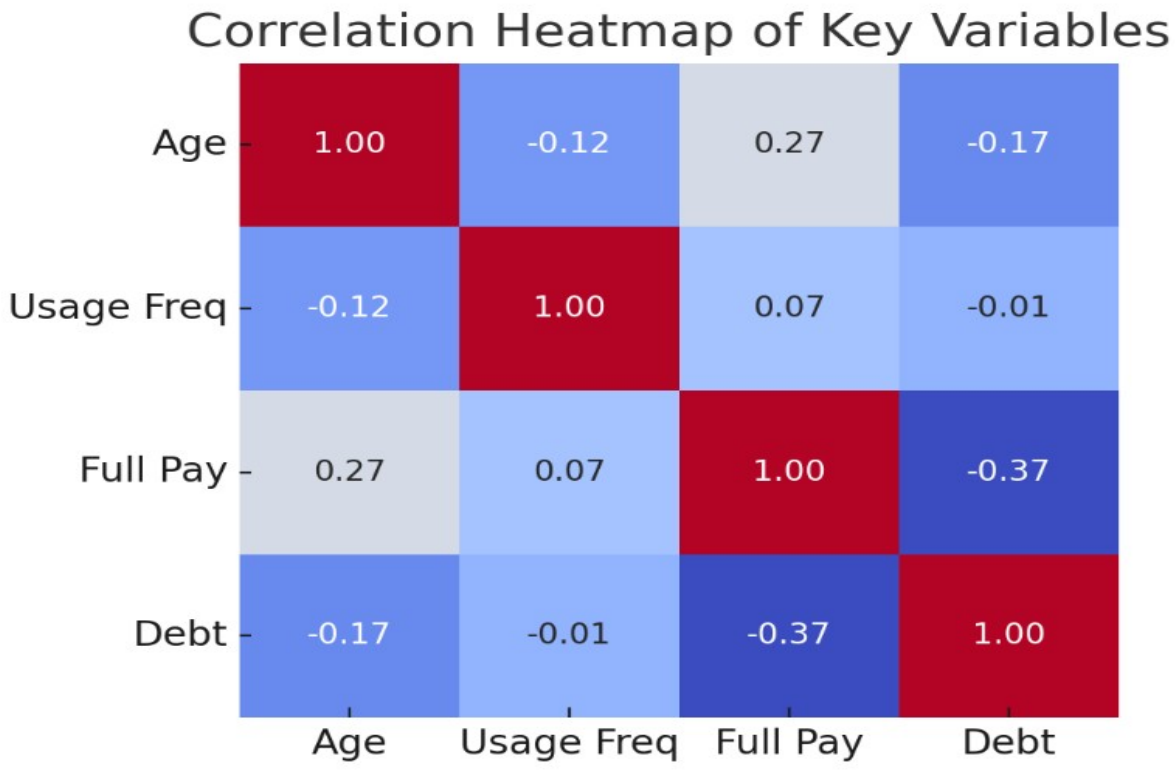


Figure 2. Correlation heatmap of key variables (Age, Usage Frequency, Full Payment behaviour, and whether the person faced credit card debt). Warmer colors (red) indicate

positive correlations while cooler colors (blue) indicate negative correlations. Notably, Age and Full Payment show a positive correlation (older individuals more likely to pay in full), whereas Full Payment and Debt have a significant negative correlation.

From **Figure 2**, we observe that **Age has a mild positive correlation with full payment behaviour** ($r \approx +0.27$). This suggests older respondents tend to more consistently pay their balances in full each cycle, aligning with the idea that financial discipline improves with age or that older individuals have more stable finances allowing full payments. Age is negatively correlated with the incidence of debt ($r \approx -0.17$), hinting that younger participants might be more prone to debt issues. The strongest correlation in the plot is a **negative correlation between "Pay in Full" behaviour and having Debt** ($r \approx -0.37$). This is intuitive and expected: those who regularly pay off their card (avoiding carrying a balance) are significantly less likely to end up in debt. In other words, habitual full payers rarely face debt problems, whereas those who don't pay in full are the ones who accumulate debt. The correlation between usage frequency and debt is nearly zero in this sample (r around –0.01), implying that how often someone uses their card is not directly linked to whether they fall into debt; what matters more is how they manage repayments. It's interesting that usage frequency has a slight negative correlation with age ($r \approx -0.12$), perhaps indicating that the youngest group includes many daily users (enjoying the ease of digital payments), whereas some older users might use weekly or monthly for more planned purchases. However, that correlation is weak.

These patterns from EDA already suggest a story: **older respondents (40–50)** appear more cautious or capable in managing credit (more frequent full payments, less debt), while **younger respondents (20–30)**, despite using cards frequently and enjoying rewards, may be at greater risk of not paying in full and thus incurring debt. Respondents in the **30–40** group seem to be of particular interest – as we will see, they combine high usage with somewhat less disciplined repayment, possibly making them the most debt-susceptible group. To confirm these age-related differences, we perform a focused analysis on age and debt next.

Age and Debt Accumulation (Chi-Square Test)

A key question of this research is whether **credit card usage leads to debt accumulation among different age groups** – essentially, is the likelihood of falling into credit card debt independent of age, or is it associated with one's age group? To investigate this, we tabulated the number of respondents in each age group who reported having faced credit card debt issues versus those who did not.

Table 1 presents both the observed frequencies from our sample and the expected frequencies if age and debt were unrelated (under the null hypothesis of independence). We also conducted a Chi-square test of independence on this contingency table.

Table 1: Observed vs. Expected Frequencies of Credit Card Debt by Age Group (Chi-Square Analysis)

Age Group	Observed # with No Debt	Observed # with Debt	Expected # with No Debt	Expected # with Debt
20–30	33	8	32.26	8.74

30–40	23	18	32.26	8.74
40–50	51	3	42.49	11.51

From the observed counts: in the **20–30** group, 8 out of 41 respondents (≈19.5%) have faced credit debt issues; in the **30–40** group, 18 out of 41 (≈43.9%) have faced debt issues; in the **40–50** group, only 3 out of 54 (≈5.6%) have had debt issues. The expected counts (if age didn't matter) would have been roughly equal proportions across age groups (given 29 total "Debt" cases in 136 sample, each group would be expected to contribute about 8–11 of those if distributed evenly). Clearly, the 30–40 age bracket shows a far higher observed debt incidence (18) than expected (~8.7), whereas the 40–50 bracket shows a far lower observed incidence (3) than expected (~11.5). The 20–30 group is very close to expected (~8 observed vs 8.74 expected).

The Chi-square test yields a test statistic **$\chi^2 = 20.54$** with **2 degrees of freedom**, and a p-value of **$p = 3.46 \times 10^{-5}$** (approximately 0.0000346). This p-value is far below any conventional significance level (0.05 or even 0.001), allowing us to **reject the null hypothesis** of independencefile-tf8a18be4ehzx1djbg3swpfile-tf8a18be4ehzx1djbg3swp. In plain terms, there is a statistically significant association between a respondent's age group and their likelihood of experiencing credit card debt issues. Age group and debt occurrence are not independent.

Looking at which cells contribute most to this χ^2 (which we can infer from the disparity between observed and expected frequencies): the **30–40 age group with Debt** is the standout contributor (Observed 18 vs Expected ~8.7). This indicates that individuals in their 30s are markedly over-represented among those who have had debt problems. Conversely, the **40–50 age group with Debt** has a very low observed count (3 vs Expected ~11.5), showing that those in their 40s are under-represented among the debt-troubled. The 20–30 group's observed is close to expected, indicating their debt incidence is about proportionate to their share of the sample.

To visualize this contrast, **Figure 1** illustrates the percentage of respondents in each age category who have faced credit card debt issues:

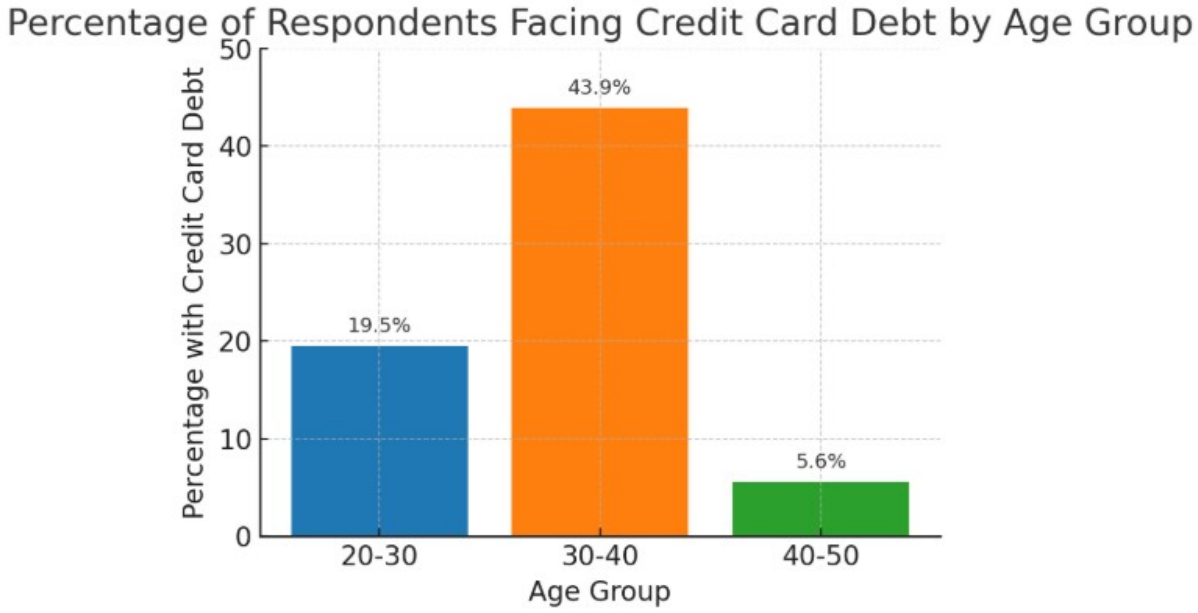


Figure 1. Percentage of respondents facing credit card debt by age group. The 30–40 age group has the highest proportion with debt (about 44%), compared to ~20% in the 20–30 group and only ~6% in the 40–50 group.

As shown in Figure 1, about **44% of individuals aged 30–40** in our sample reported having encountered credit card debt difficulties. This is more than double the rate of the younger 20–30 group (around 20%) and almost an order of magnitude higher than the oldest 40–50 group (around 5.6%). This stark pattern suggests that people in their 30s are the most vulnerable to credit card debt accumulation, at least within the scope of our survey.

Several interpretations emerge from this finding. The 30–40 age range often corresponds to a life stage with increased financial responsibilities – many in this group may be dealing with mortgages or home loans, car loans, family expenses (raising children), and so forth. They might use credit cards to juggle these expenses and occasionally overstretch. They also might not yet have the higher incomes, or savings buffers that older individuals have developed, making them more susceptible to falling behind on payments. By contrast, the 40–50 age group appears to manage credit cards extremely well: only 3 out of 54 had debt issues, which aligns with the earlier observation that they are most likely to pay in full and use cards in a controlled manner. The older group may have more financial experience and stability (and possibly higher incomes at the peak of their careers), enabling them to avoid debt traps even if they use cards frequently.

The **20–30 group** has a moderate incidence of debt issues. While many younger users are new to credit, and some do slip into debt (one in five in our sample), it is possible that the relatively lower rate compared to the 30–40 group is because younger people might have lower credit limits or fewer cards (thus a limited capacity to accumulate debt), or they might have parental guidance or support in bailing out if they get into trouble. It's also conceivable that some 20–30

individuals have not had enough time to fall into serious debt – they might still be in a cautious phase of credit use.

The Chi-square analysis thus confirms: **Age is a significant factor** in credit card debt outcomes in this sample. Specifically, being in the 30–40 age bracket is associated with a much higher risk of debt, whereas being in the 40–50 bracket is associated with a much lower risk (and 20–30 is intermediate). This finding answers our primary question in the affirmative — credit card usage and its debt consequences do differ by age group.

Sentiment Analysis of Open-Ended Responses

To complement the quantitative findings, we analyzed respondents' open-ended comments about whether and how credit card usage influenced their spending behaviour. The sentiment analysis of these comments provides insight into the **psychological and emotional side** of credit card use, which often underlies financial outcomes.

Overall, the sentiment of responses was skewed towards the negative side. Many users expressed *concern* or *acknowledged negative tendencies* related to credit card use. Common themes included overspending, impulsive purchases, and the ease of losing track of expenses when not dealing in cash. For instance, one respondent noted, *"Yes, it [credit card] may lead to over-spending and unnecessary expenses. You don't feel the pinch immediately, so you might exceed your budget."* Such sentiments were typical, reflecting wariness of credit cards as a double-edged sword.

We quantified each response with a polarity score. The distribution of these scores showed a left skew (toward the negative). Very few responses had strongly positive sentiments. The bulk of comments were either somewhat negative or neutral/mixed. Many respondents gave nuanced answers acknowledging both pros and cons (e.g., *"I enjoy the rewards, but I have to be careful because it's easy to overspend"*, which might come out neutral if the positive and negative sentiments balance).

We then calculated the **average sentiment score by age group** to see if the age patterns align with the debt findings. **Figure 3** presents these average sentiment scores for each age category:

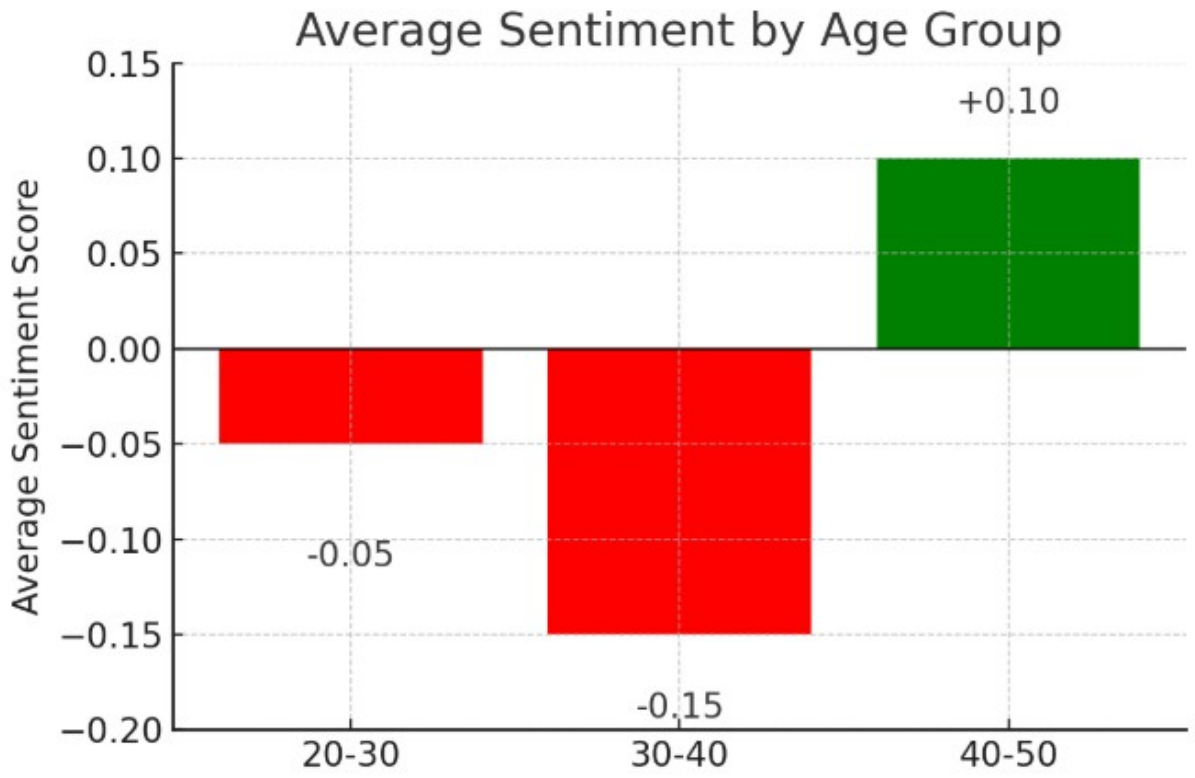


Figure 3. Average sentiment score of credit card's influence on spending behaviour, by age group. Scores range from -1 (very negative) to +1 (very positive). The 30–40 group has the most negative average sentiment (approx -0.15), the 20–30 group is slightly negative (~ -0.05), and the 40–50 group is slightly positive (~ +0.10).

We observe that the **30–40 age group indeed had the most negative average sentiment** (around –0.15 on the scale we derived). This indicates that, consistent with their higher incidence of debt, those in their 30s are also the most likely to say that credit cards negatively influence their spending. Many in this group wrote about struggling to control spending or feeling that the ease of credit led them to purchases they later found problematic. There were phrases like "tends to overspend" or explicit admissions of accumulating debt due to spending beyond means. This convergence of negative sentiment and actual debt problems reinforces the narrative: people in their 30s not only are getting into debt more, but they *feel* and recognize the problematic influence of credit cards on their behaviour.

The **20–30 group's sentiment** was on average slightly negative (just below zero, around –0.05). Many younger respondents had mixed feelings. A number said something akin to *"Yes, it influences me to spend more, but I keep an eye on it"* or *"It's convenient but I try not to let it affect my budget."* Some younger users explicitly said credit cards *helped* them manage expenses or build credit, which is a positive sentiment, but others admitted to temptation to spend beyond budget (negative sentiment). The net result is a mildly negative tilt. It suggests younger users are somewhat cautious or at least aware of the risks, which might correlate with why only 20% ended up in debt – they are learning and trying to manage responsibly, though not without occasional slips.

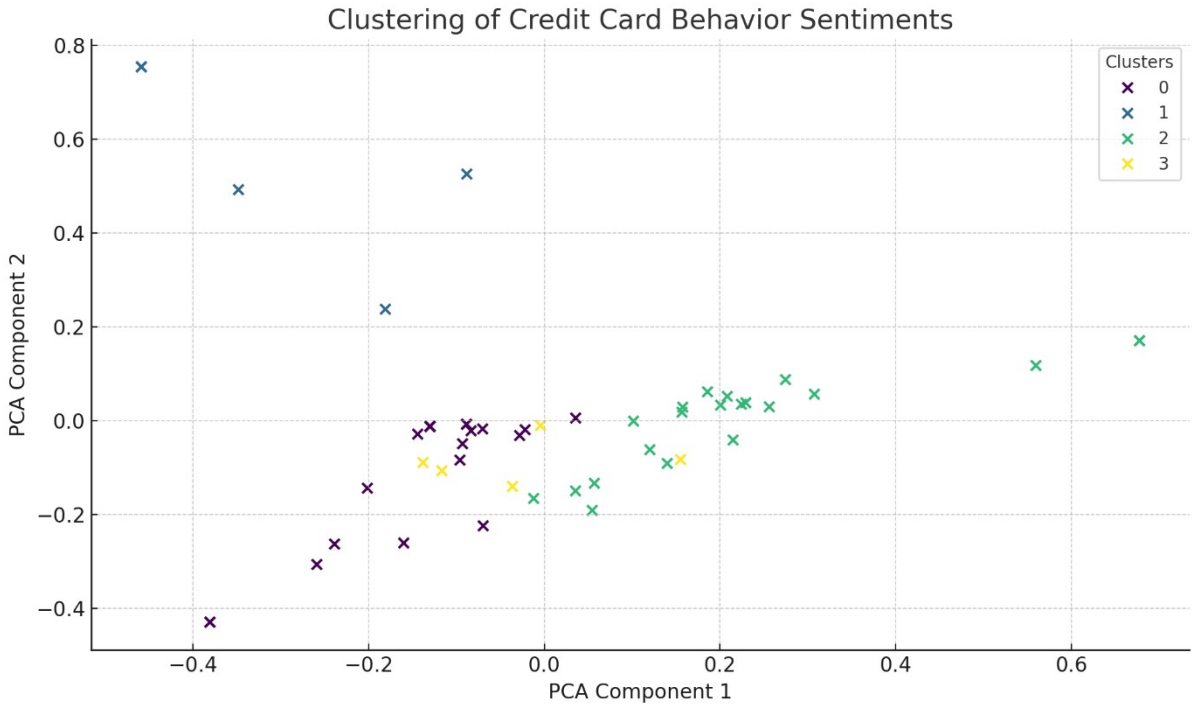
Interestingly, the **40–50 group's average sentiment was neutral to slightly positive** (around +0.10 in our analysis). Many older respondents indicated that credit cards did *not* adversely affect their spending. For example, one person in the 40–50 bracket said, *"Credit Card eases my spending as I'm fully aware of how much I can spend."* Another noted, *"I feel safe and comfortable as I don't have to keep cash, and I pay it off each month, so it doesn't make me spend more than I should."* These kinds of responses reflect a more controlled and positive relationship with credit cards – seeing them as tools of convenience and safety, rather than gateways to debt. A few even expressed positively that credit cards *helped* them in emergencies or that the rewards were a bonus on spending they would do anyway, hence no negative effect on their finances. Such perspectives explain why this group has the lowest actual debt incidence: they generally don't let the card change their spending habits adversely.

Linking sentiment to actual debt: We find a meaningful alignment – the group that is most at risk (30–40) is most negative about credit card influence, presumably from experience, whereas the most secure group (40–50) is relatively positive or neutral, possibly because they use credit cards in a controlled manner that doesn't harm them. This alignment supports the idea that **psychological factors and self-awareness go hand-in-hand with financial outcomes**. Those who acknowledged "I overspend with a card" often were the ones who had overspent and incurred debt. Meanwhile, those who said "I use it but remain in control" had indeed remained out of debt.

Beyond averages, the content of responses can be grouped into a few thematic clusters, which we explicitly did using clustering analysis.

Clustering of Credit Card Behaviour Sentiments

Using TF-IDF and KMeans clustering on the open-ended responses, we identified **four distinct clusters** representing types of attitudes/behaviours regarding credit card influence:



- **Cluster 0 – "Controlled Awareness":** This cluster consisted of responses where the respondent acknowledged the influence of credit cards but in a *controlled or positive* light. These users often said they were aware of their spending limits and used the card as a tool without falling prey to overspending. For example, a representative quote included *"I'm fully aware of how much I can spend"*. The sentiment here was generally positive or neutral. These respondents treat the credit card as an instrument of convenience and perhaps slight flexibility in timing (deferred payment), but they monitor their expenses closely. This cluster aligns mostly with older, financially mature users (many from the 40–50 group fell here). It reflects *cautious optimism* – enjoying the benefits of the card while staying self-disciplined.

- **Cluster 1 – "Overspenders":** Responses in this cluster were very terse and pointed, often explicitly using the word "overspend" or similar. Examples were simply *"Overspending"* or *"Tends to make overspending"* as entire answers. These respondents directly credit (or blame) the credit card for causing them to spend more than they should. The brevity and directness suggest a keen awareness of a problem. The sentiment is clearly negative, and this cluster correlates strongly with those who have had debt issues or at least close calls. We saw that many cluster 1 members were in the 30–40 age group, which makes sense as they had the highest debt incidence. We label this group as overspenders, and indeed this represents a risk segment for credit card issuers (and a concern for financial educators): these users readily admit the card leads them astray financially.

- **Cluster 2 – "Behavioural Influence & Regret":** This cluster's responses were also negative in sentiment but tended to be a bit more reflective or descriptive than cluster 1. People in this group wrote that credit cards *do* influence their spending behaviour, often saying "Yes, big time" or explaining a mechanism like *"people think they have more to*

spend, so they start spending more…" or *"it may lead to unnecessary expenses."* These comments show an understanding of the psychology: having a line of credit increases perceived available money, which can lead to purchasing things one wouldn't otherwise. There is a tone of regret or concern, as if the respondents are cautioning themselves (and by extension others). Unlike cluster 1's one-word answers, cluster 2 responses are slightly broader in acknowledging the effect. This cluster had a mix of ages but still skewed younger than 40. We interpret this cluster as individuals who *recognize the psychological pull* of credit cards toward overspending. They might not be as deeply in trouble as cluster 1 (some might not have actual debt yet) but they are wary and possibly have had some close calls or minor regrets.

- **Cluster 3 – "Opportunistic Users":** This cluster captured a different angle – those who might spend *more* because of credit card rewards or benefits, but intentionally so. For instance, responses included: *"Yes, if you take travel for example, I try booking tickets on specific days to get offers… hence end up spending more"* and *"I plan to travel more when I avail travel benefits and hence end up spending more. To get points."* These users are influenced by the incentives of the card (travel points, discount days, etc.) and will adjust their spending behaviour to capitalize on those opportunities, potentially spending more in certain categories than they otherwise would. The sentiment here isn't exactly regretful; it's more like a strategic admission: they know they're spending more, but it's for rewards they value. This cluster likely includes savvy users who chase rewards (some might be younger, excited by travel and points, and some could be older who have the means to utilize many benefits). It's a *risky strategy* if not managed, but not inherently negative-feeling for them – they might be happy to get those points, if they can afford the spending.

These clusters provide a nuanced view: not every user who spends more due to credit cards sees it as a bad thing (cluster 3), and not everyone avoids overspending (clusters 1 and 2 highlight those who do overspend and often regret it). Cluster 0 underscores that a segment of users remains disciplined and in control.

Relating clusters back to age and debt: Cluster 0 (controlled) was dominated by older, no-debt users; Clusters 1 and 2 (overspenders and regretful influencers) had a large overlap with the 30–40 group and those with debt issues; Cluster 3 (opportunists) was mixed but included some 20–30 and 30–40 who had not necessarily fallen into debt but could be at risk if their "strategic extra spending" goes too far.

In summary, the **textual analysis supports the quantitative findings**: many mid-life users are struggling with overspending (as shown by both their words and their actual debt incidence), whereas older users often express a controlled approach, and younger users are somewhat in-between (aware of risks, some falling prey, some actively playing the rewards game).

Predictive Modelling of Debt Risk

To further understand the drivers of debt accumulation and to test how well we could predict who is at risk, we built classification models using Age Group, Usage Frequency, and Full Payment Behaviour as predictors of the binary outcome "Faced Credit Card Debt (Yes/No)".

Our goal was not only to predict but also to interpret which factors are most important in distinguishing those who fell into debt from those who did not.

We trained logistic regression, SVM (with RBF kernel), and random forest models. **Table 2** summarizes the performance of the two most informative models (Logistic and Random Forest) as well as SVM, focusing on accuracy and the recall and F1-score for the positive class (debt = Yes). We highlight the challenge of identifying the minority class (debt cases).

Table 2: Classification Model Performance for Predicting Credit Card Debt

Model	**Accuracy**	**Recall (Debt=Yes)**	**F1-score (Debt=Yes)**
Logistic Regression	75.6%	10%	0.17
SVM (RBF Kernel)	78.0%	10%	0.18
Random Forest	75.6%	40%	0.44

All three models achieved roughly **75–78% accuracy**, which at first glance seems decent. However, since only ~21% of individuals had debt issues, a naive model that predicted "No Debt" for everyone would be about 79% accurate. Thus, accuracy alone is not a good indicator of success here – we need to look at how well the models identify the positive debt cases.

The **Logistic Regression** and **SVM** models had very low recall for the debt class (~10%). In fact, in our test sample, these models only correctly identified about 1 in 10 people who truly had debt issues. To contextualize, one logistic model confusion matrix (for a test portion) showed that it predicted "debt = Yes" for only a couple of individuals, correctly flagging 1 person who had debt but missing 9 others who did (classifying them as "No Debt"). This results in a low F1-score (~0.17–0.18) for the positive class, indicating poor balance between precision and recall on that class. Essentially, logistic and SVM were **conservative in flagging debt risk** – they tended to classify most people as not having debt (perhaps because the majority of training data had no debt), and thus they achieved high precision (when they say someone has debt, it's likely correct, but they say it for very few people) but very low recall.

Why did these models struggle? One reason is the class imbalance – with far fewer "Yes" cases, they can optimize accuracy by focusing on predicting the dominant class correctly. Another reason might be that the variables chosen, while correlated with debt, have overlap among classes (for example, not everyone in 30–40 with sometimes not full payment ended up in debt; some managed to escape, so the models find it hard to draw a perfect line).

In contrast, the **Random Forest model** performed better in identifying debt cases. It achieved a recall of 40% for the debt class, meaning it correctly identified 4 out of 10 individuals who had debt issues. Its precision was lower than logistic's, but overall, it struck a more useful balance (F1 ~0.44 for debt class). The random forest was able to pick up more subtle combinations of factors by leveraging its ensemble of decision trees. For instance, it might identify subsets like "age 30–40 AND does not always pay in full" as high risk, and flag those. Indeed, the random forest emerged as the best model for capturing those at risk of debt while still maintaining a reasonable overall accuracy (75.6%).

It's worth noting that even the random forest missed 60% of the debt cases (recall 40%), which indicates that predicting financial distress is challenging with just these behavioural indicators. There may be other unmeasured factors (like income, sudden life events, etc.) that also affect debt outcomes. Nonetheless, among our models, the random forest gives the most confidence in identifying a meaningful fraction of at-risk individuals.

Looking at feature influence: The **logistic regression coefficients** provide insight into how each factor influences the likelihood of debt (holding others constant). The logistic model's coefficients (in standardized form) were approximately: **Age Group = –0.50**, **Usage Frequency = –0.42**, **Full Payment Behaviour = –0.52**. All three are negative, which aligns with our earlier findings – an increase in any of these (i.e., being older, using more frequently, or paying in full more often) correlates with a decreased probability of having debt. In particular, "Full Payment" had the largest magnitude, suggesting that not paying in full is the strongest predictor of debt problems (which makes sense; it directly leads to revolving debt). Age also is a significant predictor: moving from a younger category to an older one reduces debt risk (the model quantifies it, roughly, older respondents have significantly lower odds of debt after controlling for usage and payment behaviour). Interestingly, higher usage frequency had a slight protective factor (negative coefficient) in the logistic model when controlling for the other two factors – this is a bit counter-intuitive but likely reflects that some frequent users are older and disciplined (so frequency alone isn't causing debt; it only leads to debt if paired with not paying in full). In simpler terms, **not paying the full balance** is the strongest risk factor, followed by being **in the 30–40 age group** (as opposed to older), whereas **using the card frequently per se did not add much risk** if one paid it off responsibly.

The decision trees in the random forest also indicated feature importance in line with this: typically, whether someone pays in full or not was the top split, and age was also frequently used in splits. Usage frequency was of lesser importance in the splits, again underscoring that frequency by itself isn't as decisive as payment behaviour and age.

In summary, the predictive modelling reinforces our understanding:

- **Full Payment Behaviour** is the most critical factor – those who do not always pay in full were far more likely to end up in debt (this factor alone goes a long way in separating risky vs. safe users).
- **Age** plays a substantial role – being in the 30–40 group markedly increases risk relative to being older, even when controlling for payment habits and usage. The models treat age 20–30 also as higher risk than 40–50, but the 30–40 is the peak risk.
- **Usage Frequency** by itself is not a strong predictor of debt once your account for payment behaviour and age. This suggests that using your card daily isn't dangerous if you're responsible with repayment; conversely, even infrequent users can fall into debt if they only make minimum payments.

The limitations of the models (low recall in some cases) highlight that predicting financial trouble is not easy with limited variables. However, the models (especially random forest) did

correctly identify many of the 30–40, non-full-payers as debtors, which aligns with our analysis.

Conclusion and Policy Implications

This study explored the intricate relationship between age groups and credit card usage behaviour, using a combination of survey data analysis, statistical testing, machine learning modelling, and sentiment evaluation. The evidence strongly indicates that **age is a significant factor** in credit card debt accumulation. We found that individuals' behaviours and outcomes with credit cards vary by life stage in meaningful ways.

Core Findings: The Chi-square test confirmed a strong association between age group and debt incidence ($\chi^2 = 20.54$, $p < 0.001$), implying that age group is not just incidentally but statistically linked to whether a person accumulates credit card debt. Specifically, the **30–40 age group emerged as the highest-risk segment** – about 44% of respondents in their 30s reported facing credit card debt issues, compared to only ~20% of those in their 20s and an extremely low ~6% of those in their 40s. This makes the 30–40 cohort a clear outlier in terms of vulnerability to debt.

Multiple strands of our analysis converge on explaining why this is the case. Those in their 30s often exhibited "erratic" credit behaviour: they had the highest proportion of individuals who sometimes or rarely paid their balance in full (leading to revolving balances), and their own survey comments reflected concern and negative sentiment about overspending. This age group is likely juggling significant expenses (family needs, home loans, etc.) while still building financial stability, and perhaps they are tempted by reward programs or necessity to rely on credit, leading to higher debt. They may also have higher credit limits than younger consumers (as credit history builds in their 30s), enabling them to accumulate more debt if unchecked.

In contrast, the **40–50 age group demonstrated financial maturity** in credit card usage. They had the highest rate of always paying the full balance and largely used cards for planned expenses or convenience rather than extending their spending. Only 3 out of 54 in this group had debt issues, which is an exceptionally low incidence. Sentiment analysis and clustering confirmed that many in their 40s view credit cards neutrally or positively and feel in control of their spending (cluster 0: "Controlled Awareness"). It appears that by this stage in life, either through experience or better financial health, individuals manage credit cards responsibly and avoid the debt spiral.

The **20–30 age group** showed some early signs of risky behaviour (8 out of 41 had debt issues, ~20%). While not as severe as the 30s group, younger adults did have instances of overspending and a sizeable minority carrying balances. Many younger respondents are still learning to navigate credit; their sentiment was only slightly negative overall, indicating a mix of those who fell into trouble and those who are cautious. With financial education and limited exposure, many in this group could potentially avoid major debt, but they are a demographic that could go either way as they move into the next age bracket.

One of the most important behavioural factors across all ages was the **practice of paying the card balance in full**. We consistently found that those who refrain from full payments (opting

for partial payments or minimum payments) are far more likely to accumulate debt due to interest. This was evident in correlation analysis, in model feature importances, and simply in the data (most of the 29 people who had debt issues were those who did not always pay in full). Encouraging the habit of full monthly payment (or at least significantly more than the minimum) is thus a key preventive measure against debt.

Sentiment & Psychological Indicators: Another noteworthy finding is that negative sentiment about credit card influence often aligned with actual debt presence. People who expressed that credit cards caused them to overspend were frequently those who had indeed overspent and incurred debt. This suggests that self-awareness of credit misuse is present in many individuals; those in trouble often *know* the card has influenced them negatively. This opens avenues for interventions – if users can recognize the problematic influence early (as some in cluster 2 did), they might take corrective action before debt mounts. Conversely, the neutral/positive sentiment among disciplined users correlates with better outcomes, showing that a healthy psychological relationship with credit (viewing it as controlled spending) goes together with financial stability.

Policy and Stakeholder Implications: Given these findings, there are several implications for different stakeholders:

- **For Banks and Credit Card Issuers:** There is a case for designing **age-targeted credit education and management tools**. Banks could provide special counseling or alerts to customers in the 30–40 age range, knowing this group is prone to debt accumulation. For instance, issuing alerts when a customer starts to carry a balance or when their credit utilization spikes could help mid-age users reconsider their spending. Credit card issuers might also refine credit limits or promotional offerings for certain age groups – for example, ensuring younger users (20–30) aren't given limits they cannot handle, and providing budgeting features in apps that appeal to the tech-savvy youth. Since rewards and offers can encourage extra spending (our cluster 3 shows some people spend more to get rewards), issuers should accompany big promotions with reminders about responsible usage. **Financial literacy content** could be included with monthly statements or apps, especially targeting the risky behaviours (like only paying minimums). In short, banks have the data on who is carrying balances; leveraging that to guide and caution users, particularly those in their 30s, could improve customer financial health and reduce defaults.
- **For Policymakers and Educators:** The results underline the importance of **financial literacy programs tailored to life stages**. Young adults (just entering the workforce and eligible for credit cards) should be educated about the perils of only making minimum payments and the long-term cost of credit card interest. Educational initiatives in colleges or first jobs can emphasize good credit habits. For those in their 30s, who may not seek out financial education as actively, community programs or online campaigns highlighting strategies to manage multiple debts (like debt consolidation, prioritizing high-interest debt payoff) could be beneficial. Policymakers could also consider regulations that enforce clearer disclosures about how long and costly it will be to pay off a big balance with minimum payments – ensuring consumers

fully grasp the implications of revolving credit. Additionally, since our findings are contextual (with presumably many respondents in India given the references), policymaking like the RBI's caution to lenders is apt: regulators may encourage or mandate that credit card companies provide installment plan options or lower interest for those who consistently pay on time, to prevent the snowball of debt in the 30s age group. Promoting digital tools that integrate bank accounts and cards to show a user's full financial picture could help consumers in mid-life manage better.

- **For the Individuals/Researchers:** Individuals can take away that **full payment behaviour is critical** – if one cannot pay the full balance, it's a clear warning sign that spending is exceeding means. The study also suggests one reflect on one's own sentiment towards credit cards; if you find yourself often regretting credit card purchases, that's a psychological cue to change your usage pattern before it leads to financial strain. Future research could build on this work by incorporating more variables such as income level, family size, or even personality traits, to see how they interact with age in determining debt risk. A longitudinal study following individuals as they age from 20s to 40s could provide causal insights: for example, is it truly the life stage (30s) that causes the spike in debt risk, or do certain cohorts behave differently irrespective of age? Additionally, researchers could look at transaction-level data to see exactly what kinds of spending lead to debt (e.g., medical emergencies vs. luxury spending) among different ages.

Limitations

It should be noted that this study relies on self-reported survey data, which can be subject to biases. Some respondents might underreport negative behaviours (though the anonymity likely reduced that). The sample size (136) is modest and may not be fully representative of the broader population of credit card users, especially since it was skewed toward those who own cards and are literate enough to take an online survey. Also, the study was cross-sectional – capturing a snapshot in time – so when we talk about age differences, there is a possibility of cohort effects (people who are 40–50 today grew up in different economic conditions than those who are 20–30 today). We assumed differences are age-related, but some differences could be generational. Nonetheless, the patterns observed are strong enough to warrant attention. Future work could expand the sample diversity (include different income groups or geographic regions) and examine how consistent these findings are.

In conclusion, the analysis shows that **credit card usage does have different implications for debt accumulation across age groups**. The "danger zone" identified is the 30–40 age range, where many individuals are leveraging credit heavily and not always managing to repay in full, thus falling into debt. Younger users are not immune, but some have time to course-correct, while older users generally exhibit wiser credit practices. By understanding these patterns, stakeholders can implement targeted strategies to improve credit outcomes – helping consumers enjoy the benefits of credit cards (convenience, rewards, emergency support) without sliding into debt traps. The key is fostering financial habits (like timely full payments and mindful spending) early on and providing support mechanisms for those in the high-

pressure financial years of their 30s. With such measures, credit cards can be used as the helpful financial tools they're meant to be, rather than stepping stones into debt.

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